

1. Let the vector function $\vec{r}: [-\sqrt{5}, 3] \rightarrow \mathbb{R}^2$ be differentiable at -1 with $\vec{r}(-1) = (-1, \pi)$ and $\vec{r}'(-1) = (-\sqrt{2}, \frac{1}{3})$. If the function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ is differentiable at the point $(-1, \pi)$ with $f_x(-1, \pi) = 2\sqrt{2}$ and $f_y(-1, \pi) = 6$, then explain why the function $f \circ \vec{r}$ is differentiable at -1 find the derivative $(f \circ \vec{r})'(-1)$.

The function $\vec{r}$ is differentiable at -1 and the function f is differentiable $\vec{r}(-1) = (-1, \pi)$, then by the chain rule, the composition $f \circ \vec{r}$ is differentiable at -1 with

$$\begin{aligned} (f \circ \vec{r})'(-1) &= \nabla f(\vec{r}(-1)) \cdot \vec{r}'(-1) = \nabla f(-1, \pi) \cdot (-\sqrt{2}, \frac{1}{3}) \\ &= (f_x(-1, \pi), f_y(-1, \pi)) \cdot (-\sqrt{2}, \frac{1}{3}) = -f_x(-1, \pi)\sqrt{2} + \frac{1}{3}f_y(-1, \pi) \\ &= -\sqrt{2} \times 2\sqrt{2} + 6 \times \frac{1}{3} = -4 + 2 = -2. \end{aligned}$$

2. (i) $\nabla f(x, y, z) = (\cos x \cos y \tan z, -\sin x \sin y \tan z, \sin x \cos y \sec^2 z)$.
 $\nabla f(\frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}) = (\frac{3\sqrt{2}}{4}, -\frac{\sqrt{6}}{4}, \sqrt{2})$. So

$$D_{\vec{u}}f\left(\frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}\right) = \nabla f\left(\frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}\right) \cdot \vec{u} = \left(\frac{3\sqrt{2}}{4}, -\frac{\sqrt{6}}{4}, \sqrt{2}\right) \cdot \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) = \frac{7}{4}$$

- (ii) $\nabla f(x, y, z) = (4x - 2yz + 1, -2y - 2xz + 3, 3z^2 - 2xy)$; $\nabla f(1, 1, 1) = (3, -1, 1)$. Now we need the direction vector which is the unit vector in the direction of $\vec{r}'(-\frac{\pi}{4})$.
 $\vec{r}'(t) = (-\sin t, -\cos t, 1)$. So, $\vec{r}'(-\frac{\pi}{4}) = (\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 1)$. The unit vector is then $\vec{u} = (\frac{1}{2}, -\frac{1}{2}, \frac{1}{\sqrt{2}})$. Thus the directional derivative is

$$D_{\vec{u}}f(1, 1, 1) = \nabla f(1, 1, 1) \cdot \vec{u} = (3, -1, 1) \cdot \left(\frac{1}{2}, -\frac{1}{2}, \frac{1}{\sqrt{2}}\right) = \frac{3}{2} + \frac{1}{2} + \frac{1}{\sqrt{2}} = 2 + \frac{1}{\sqrt{2}}.$$

3. (i) We seek (x, y) such that $\nabla f(x, y) \cdot \vec{u} = 1$.

$$\nabla f(x, y) = (4x, 1 + 6y).$$

So,

$$\nabla f(x, y) \cdot \vec{u} = 1 \Leftrightarrow (4x, 1 + 6y) \cdot \left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right) = 1 \Leftrightarrow 4x + 12y = \sqrt{5} - 2.$$

(ii) We seek $\mathbf{u} = (x, y)$ with $x^2 + y^2 = 1$ such that $\nabla g(a, b) \cdot \mathbf{u} = 0$. We get

$$\begin{cases} 2x + 3y = 0 \\ x^2 + y^2 = 1 \end{cases} \Rightarrow \mathbf{u}_1 = \left(\frac{3}{\sqrt{13}}, -\frac{2}{\sqrt{13}} \right) \quad \text{and} \quad \mathbf{u}_2 = \left(-\frac{3}{\sqrt{13}}, \frac{2}{\sqrt{13}} \right)$$

4. Let $\vec{u} = \frac{1}{\sqrt{5}}\langle 2, 1 \rangle$ and $\vec{v} = \frac{1}{\sqrt{2}}\langle 1, -1 \rangle$. If the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is differentiable at the point (a, b) with $f_{\vec{u}}(a, b) = -2$ and $f_{\vec{v}}(a, b) = 1$, then find:

(i) $\nabla f(a, b)$;

We recall that $\nabla f(a, b) = \langle f_x(a, b), f_y(a, b) \rangle$ and that $f_{\vec{u}}(a, b) = \nabla f(a, b) \cdot \vec{u}$, $f_{\vec{v}}(a, b) = \nabla f(a, b) \cdot \vec{v}$. Therefore, we obtain

$$\begin{cases} f_{\vec{u}}(a, b) = \nabla f(a, b) \cdot \vec{u} = -2, \\ f_{\vec{v}}(a, b) = \nabla f(a, b) \cdot \vec{v} = 1 \end{cases} \Leftrightarrow \begin{cases} 2f_x(a, b) + f_y(a, b) = -2\sqrt{5} \\ f_x(a, b) - f_y(a, b) = \sqrt{2} \end{cases}$$

from which we find that

$$f_x(a, b) = \frac{-2\sqrt{5} + \sqrt{2}}{3} \quad \text{and} \quad f_y(a, b) = \frac{-2\sqrt{5} - 2\sqrt{2}}{3}.$$

Therefore,

$$\boxed{\nabla f(a, b) = \left(\frac{-2\sqrt{5} + \sqrt{2}}{3}, \frac{-2\sqrt{5} - 2\sqrt{2}}{3} \right)}.$$

(ii) The maximum increase of the function f at the point (a, b) is given by

$$\begin{aligned} \|\nabla f(a, b)\| &= \sqrt{(f_x(a, b))^2 + (f_y(a, b))^2} \\ &= \frac{1}{3} \sqrt{(-2\sqrt{5} + \sqrt{2})^2 + (-2\sqrt{5} - 2\sqrt{2})^2} = \frac{1}{3} \sqrt{50 + 4\sqrt{10}}. \end{aligned}$$

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